Vibration Isolator – Maintenance Practices

1. Removal/Installation

Refer to Figure 1 for details.

A. Remove Vibration Isolator

1) Remove engine cowlings.

2) Support engine weight.

3) Remove engine to mount attachment bolt and washer, using wrench SD12524. Remove barrel nut and lock segment.

4) Cut lockwire and remove four bolts and washers securing vibration isolator to engine. On top isolator only, remove engine slinging bracket with the two front securing bolts.

5) Remove vibration isolator and gasket.

6) Check vibration isolator for deformation, overhaul, or replacement. (Refer to Para 3.A.1)).

Note Change complete vibration isolator engine mount or partial, as required in accordance with Para 3., Inspection/Check.

B. Install Vibration Isolator

1) Install countersunk washer under engine to mount attachment bolt head and insert bolt in vibration isolator. Ensure recess in vibration isolator is towards nacelle structure attachment point.

2) Check that mounting faces are clean and undamaged.

3) Position gasket (supplied with vibration isolator) and vibration isolator on engine mounting face.

4) Install four vibration isolator retaining bolts and washers. Torque to 200 to 250 pound–inches and lockwire.

Note Install four new vibration isolator retaining bolts at each engine change (refer to Table 1 — Hardware List to Attach Engine Vibration Mount to Nacelle and Engine).

On top isolator only, the two front securing bolts are longer and also secure the rear engine slinging bracket. One washer is installed between the slinging bracket and the isolator and one washer under the head of each bolt.

WARNING WEAR RUBBER GLOVES, EYE PROTECTION, AND A RESPIRATOR WHEN HANDLING SOLVENTS, SEALING COMPOUNDS, OR ADHESIVES. CONTACT WITH SOLVENTS, SEALING COMPOUNDS, OR ADHESIVES AND THEIR VAPOURS MAY CAUSE IRRITATION TO THE SKIN, THE EYES, AND THE LUNGS. FAILURE TO WEAR PROPER PROTECTION MAY CAUSE INJURIES, SERIOUS ILLNESS, OR DEATH TO PERSONNEL. OBEY ALL HEALTH AND SAFETY PRECAUTIONS AND REGULATIONS CONCERNING STORAGE, USE, AND DISPOSAL.

5) Apply corrosion inhibiting compound (CIC) MASTINOX 6856K or CA-1000 to the face and inside diameter of each engine mount adapter bushing, shank of engine attachment bolt and washers.

Note Do not apply any corrosion inhibiting compound to the engine attachment bolt threads or vibration isolator elastomers.

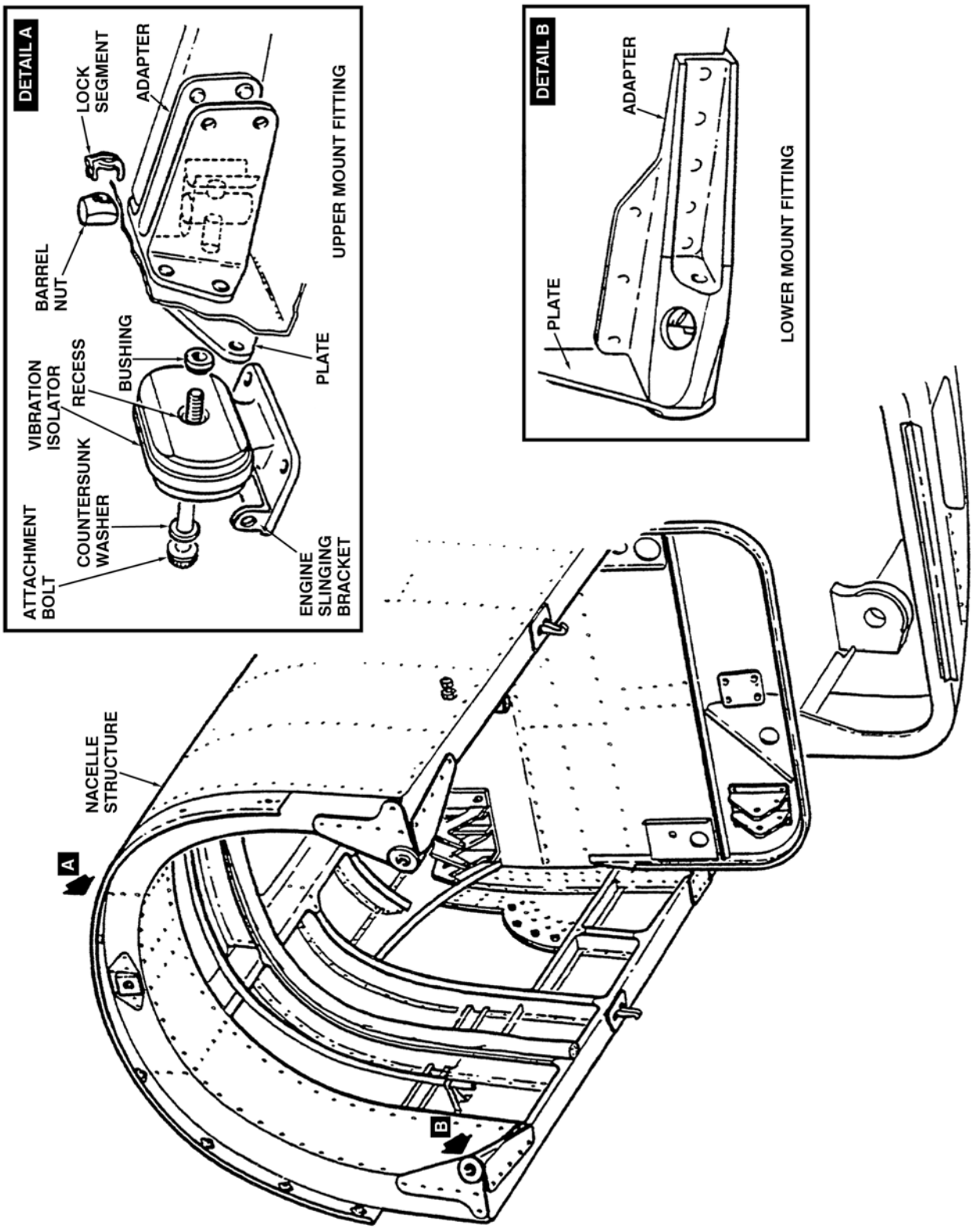
6) Secure engine attachment bolt to barrel nut and lock segment, using wrench SD12524. Torque bolt to 325 to 375 pound–inches.

7) Remove engine sling.

8) Install cowlings.

Table 1 Hardware List to Attach Engine Vibration Mount to Nacelle and Engine

QTY PER ENGINE	DESCRIPTION	400 P/N
3	BOLT – VIBRATION ISOLATOR ASSY TO NACELLE	NASM21250–06042
3	WASHER – VIBRATION ISOLATOR ASSY TO NACELLE	NASM20002C–6
3	NUT – BARREL, VIBRATION ISOLATOR ASSY TO NACELLE	B42–624
2	FOR TOP MOUNT ONLY REAR BOLTS – SPECIAL, VIBRATION ISOLATOR ASSY TO ENGINE CASE	CSP282–29
	AND	
2	FRONT BOLTS – VIBRATION ISOLATOR ASSY TO ENGINE CASE	AN6H6A
8	FOR SIDE MOUNTS BOLT – SPECIAL, VIBRATION ISOLATOR ASSY TO ENGINE CASE	CP282–29
14	WITH MOD 6/1706 WASHER – FLAT, VIBRATION ISOLATOR ASSY TO ENGINE CASE	NAS1149C0663R
3	GASKET – VIBRATION ISOLATOR ASSY TO ENGINE CASE	R20802–2



Engine Mounting
Figure 1

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2. Disassembly/Assembly

Refer to Figure 2 for details.

A. Disassemble Engine Vibration Isolator Assembly

1) The core (2) is press fit on stud (3). Pull core off stud by hand or use soft mallet to tap housing away from core.

2) Remove core (8) and stud (3) from housing (4).

3) Remove bumper tube (6) from stud (3).

CAUTION **WHEN DRILLING, BE CAREFUL NOT TO DAMAGE CORE (8). CORES DAMAGED DURING DRILLING MUST BE REPLACED.**

4) Hold protruding end of stud (3) and drill out flared portion of stud (3) in core (8) and discard stud.

5) Remove and discard gasket (5).

6) Visually inspect cores for damage.

Note Core assemblies removed from an engine vibration isolator are to be retained as a matched pair, visually inspected and re-installed as a matched pair.

7) If one core has visible damage, replace both cores as a pair.

Note If engine vibration isolator housing does not pass the inspection requirement, replace the entire engine vibration isolator assembly and associated hardware as shown in Table 2 — Component List for Figure 2.

B. Assemble the Engine Vibration Isolator

1) Insert small end of stud (3) into core (8).

a) Flare the stud in the countersink of core (8) with 100° flare tool.

Note Flare stud only enough to hold core so it will not come off the stud. Too much flare could crack the stud. The deep countersink in the core is to ensure the flared end of the stud is below the surface of the core.

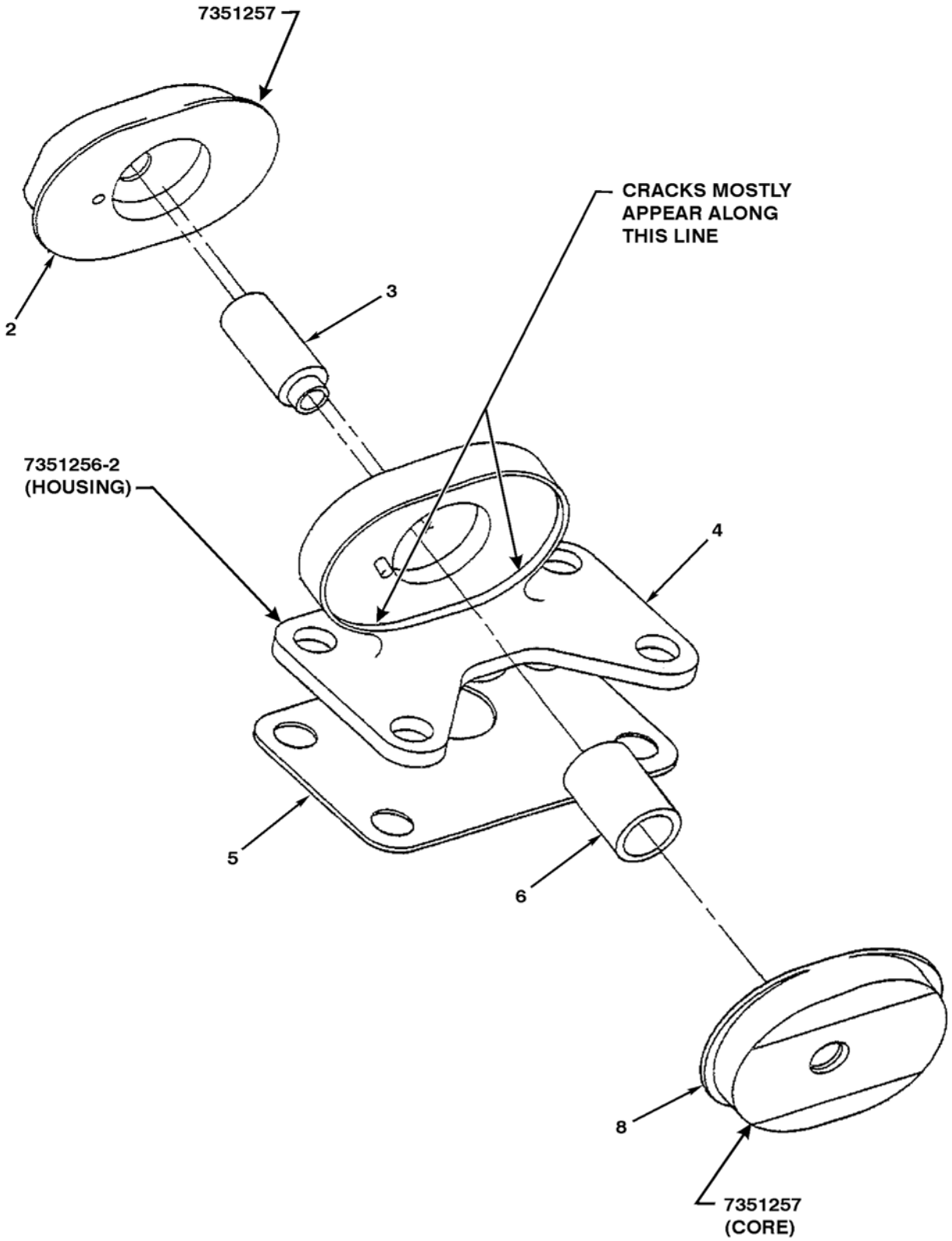
2) Install bumper tube (6) over stud and place sub-assembly on housing (4) aligning the pin on the housing with the hole in the core.

Note The spring pin used on housing P/N 7351256–2 (refer to Figure 2) and the cast pin used on the current production housing assure correct sub-assembly orientation and do not carry load. The blind hole in the current production core assemblies has been enlarged to simplify assembly. If core assemblies with a smaller diameter blind hole are to be installed on the cast pin housing, enlarge blind hole with a 5/32 inch drill to facilitate installation.

3) Press fit the core (2) onto stud (3).

4) To keep the press fit core (2) on the stud (3) for handling and storage purposes, place a plastic bonding tie strap or equivalent through the stud hole and secure it around the two cores. In some instances the press fit is not sufficient to hold the isolator together. This happens occasionally and is considered normal. The isolator can be used as is if this occurs.

5) Install new gasket (5) on housing.



Engine Vibration Isolator
Figure 2

Table 2 Component List for Figure 2

ITEM	QTY	DESCRIPTION	PART NUMBER	SERIES
-1	1	ISOLATOR ASSEMBLY	7350721–7	
-2	1	CORE	7351257–9	
-3	1	STUD	7351261–1	
-4	1	HOUSING	7351256–2	
-5	1	GASKET	R20702–2	
-6	1	TUBE, BUMPER	7351260–1	
-8	1	CORE	7351257–7	

3. Inspection/Check

A. Condition, Separation, Corrosion Cracking, Deformation and Security Check

Refer to Figure 3 for details.

1) Remove vibration isolator and check for permanent set or sag. Maximum allowable 0.125 inch.

2) Check for delamination of elastomer. Maximum allowable 1.0 inch length or 0.0625 inch depth.

3) Check for corrosion and cracking. Pay particular attention to corrosion and cracks at base of housing (P/N 7351256–2) right next to the edge of the cores (pucks).

4) Replace vibration isolator if limits in Para 3.A.1) or Para 3.A.2) are exceeded.

5) Install vibration isolator.

B. Overhaul Check

1) Inspect the engine vibration isolator in accordance with Para 3.A.

2) Inspect the engine vibration isolator housing P/N 7351256–2, in accordance with Para 3.C.

3) If vibration isolator engine mount does not pass the Fluorescent Inspection, then the engine mount must be replaced, including the associated hardware (refer to Table 1 — Hardware List to Attach Engine Vibration Mount to Nacelle and Engine).

Note Rework limits as per PSM 1-6-6A, Component Maintenance Manual (Barry Controls Type 7350721), DO NOT APPLY.

C. Fluorescent Penetrant Inspection

Refer to Figure 2 for details.

1) Function

a) This procedure documents the Fluorescent Penetrant process used to detect discontinuities on the surface of the DHC-6 Engine Vibration Isolator Housings, P/N 7351256–2.

b) Materials:

- 17–4PH Stainless Steel Casting (130–150KSI)

c) Type/Sensitivity:

- Type I, Level 3 or 4 in accordance with ASTM–E–1417

2) Applicability

This process is applicable for maintenance (in-service) inspections of the vibration isolator housing only.

3) Tools, Materials, Equipment and Personnel

a) Fluorescent Penetrant

1 Black Lights:

The black light must have a minimum power rating of 100 Watts and a minimum acceptable intensity of 1000 µWatts/cm2 measured at a distance of at least 38 cm (15 inches) from the front of the bulb or filter. A suitable model is Magnaflux 2Z–24A P/N: 513721 or equivalent. Black light reflectors and filters shall be checked for cleanliness and integrity prior to carrying out the FPI process.

2 Materials:

- Penetrant System shall be Type I, Level 3 or 4, in accordance with ASTM–E–1417.
- The materials used for the penetrant system above shall be compatible, come from the same manufacturer and be listed on QPL–AMS–2644.
- Materials shall be tested in accordance with Table 1 of ASTM–E–1417.

3 Inspection Area:

The inspection area for stationary systems for Type I penetrants shall be checked as specified in Table 1 — Hardware List to Attach Engine Vibration Mount to Nacelle and Engine and shall be clean and free from excessive fluorescent contamination and excessive visible light background in accordance with ASTM–E–1417.

4 Personnel Qualification:

Personnel doing examinations shall be qualified and certified in accordance with ASTM–E–1417, sensitivity Level 3, or to a standard acceptable to the operators cognizant airworthiness authority.

4) Procedure

a) Surface Preparation:

Thoroughly clean the inspection surface by solvent, chemical, or mechanical cleaning method in accordance with ASTM–E–1417.

b) Penetrant System:

Penetrant system shall be Type I, Level 3 or 4 in accordance with ASTM–E–1417.

1 Penetrant Application:

Penetrant shall be applied in accordance with ASTM–E–1417.

2 Penetrant Dwell Time:

Dwell time shall be 60 minutes. The drain off time shall be considered as part of the penetrant dwell time. If the method used is immersion, parts shall be immersed in penetrant for no more than one half the dwell time. The long dwell time may result in drying of the penetrant, fresh penetrant shall be reapplied periodically to avoid drying.

3 Penetrant Removal:

Penetrant shall be removed by applicable method in accordance with ASTM–E–1417.

4 Drying:

Parts shall be dried prior to the application of developer in accordance with ASTM–E–1417.

5 Developing:

Developer application shall be done with a developer listed in QPL–AMS–2644 for Type I, Level 3 or 4 in accordance with ASTM–E–1417.

6 Inspection:

Inspection of parts shall be conducted in accordance with ASTM–E–1417, in a darkened booth using black light with a minimum intensity of 1000 µWatts/cm2 on the surface of the part being inspected. The inspector shall be in the darkened booth for at least one minute prior to doing inspection so that his/her eyesight will adapt to dark viewing. The inspector shall not wear light sensitive or permanently darkened lenses while doing the inspection.

5) Inspection Criteria

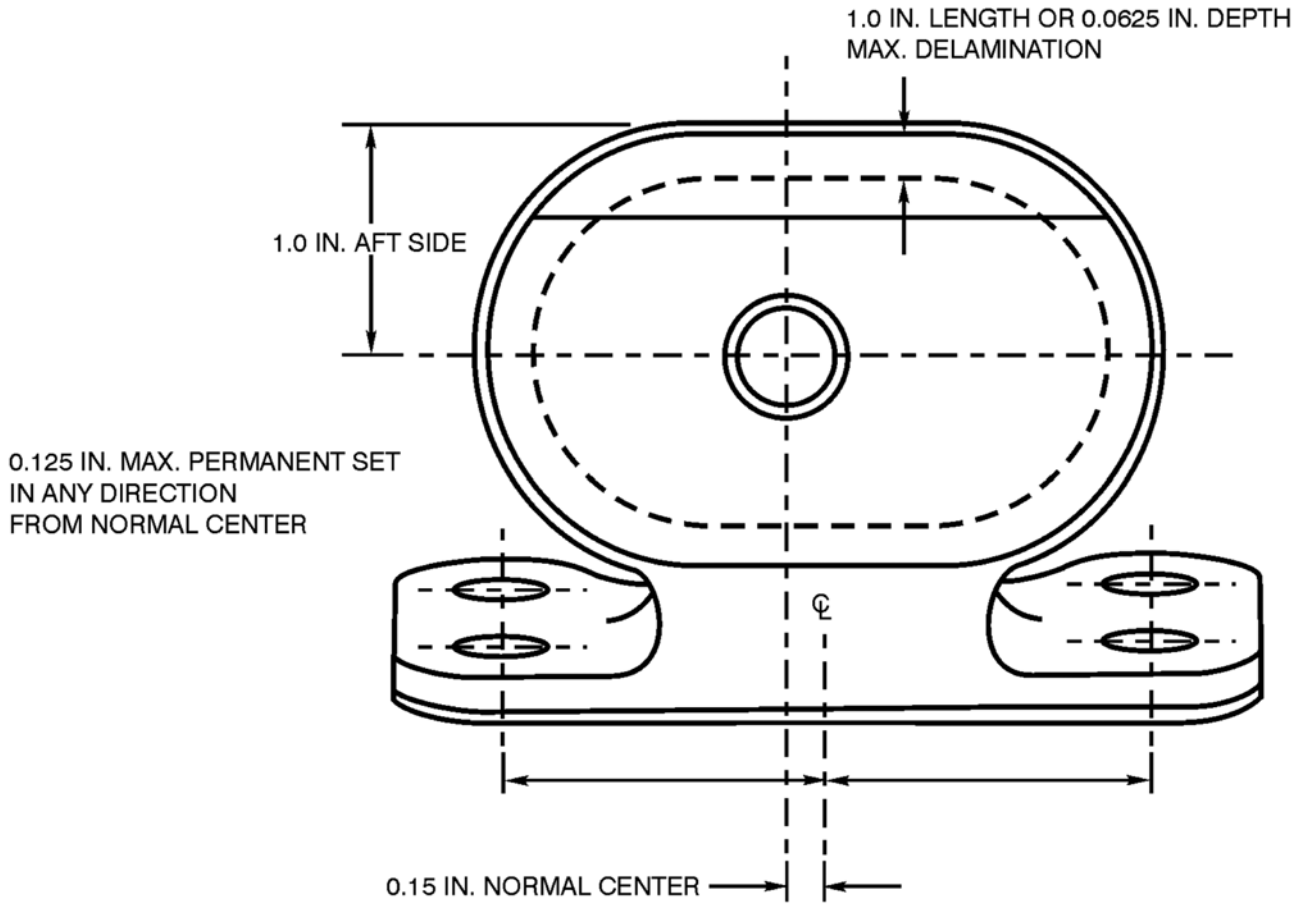
Linear indications and non-linear indications shall be interpreted. All indications interpreted to be defects shall be cause for isolation housing REJECTION.

6) Post Inspection Processes

Post-inspection cleaning and marking shall be carried out in accordance with ASTM–E–1417.

7) Inspection Records

Inspection records, at the inspection facility, shall show the part number, date of inspection, number of parts inspected, number of parts accepted and rejected, reference to any rejection documents and the identity of the inspector. Additionally, the log should record the nature, size, number and location of the rejectable indications and shall be noted on the rejection document.



Deformation Check
Figure 3

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